

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Any 3 x (1) from: Random mating (1) non disjunction does not occur/ no mutations (1) Large population (1) No immigration/ no emigration / population is isolated (1) No selection/ selective advantage (1)	3			3		
		(ii)	I II III IV	55% (1) 3025 (1) 2025 (1) 4950 (1)		4		4	4	
	(b)			Genetic drift/ founder effect (1) Emigrated (small) group with an abnormally high frequency of N / abnormally low frequency of M (1) Emigrated population did not mate with general Pennsylvanian population/ no {allele / gene} flow (1)		1	2	3		
				Question 4 total	3	5	2	10	4	